

Maddie Bird, Leisha Khapre, Inayaa Khoja, Gaurav Shatoor, Jade Winebright
Dr. Alfredo Garcia, Anshul Yadav, Arnie Tyler - MerLion Advisory Group

Problem Statement

To replace hazardous and inconsistent manual diving operations, project LiAIson develops a workflow to detect stationary anomalies using synthetic underwater point clouds, providing a safer, more reliable method for critical maritime infrastructure inspection.

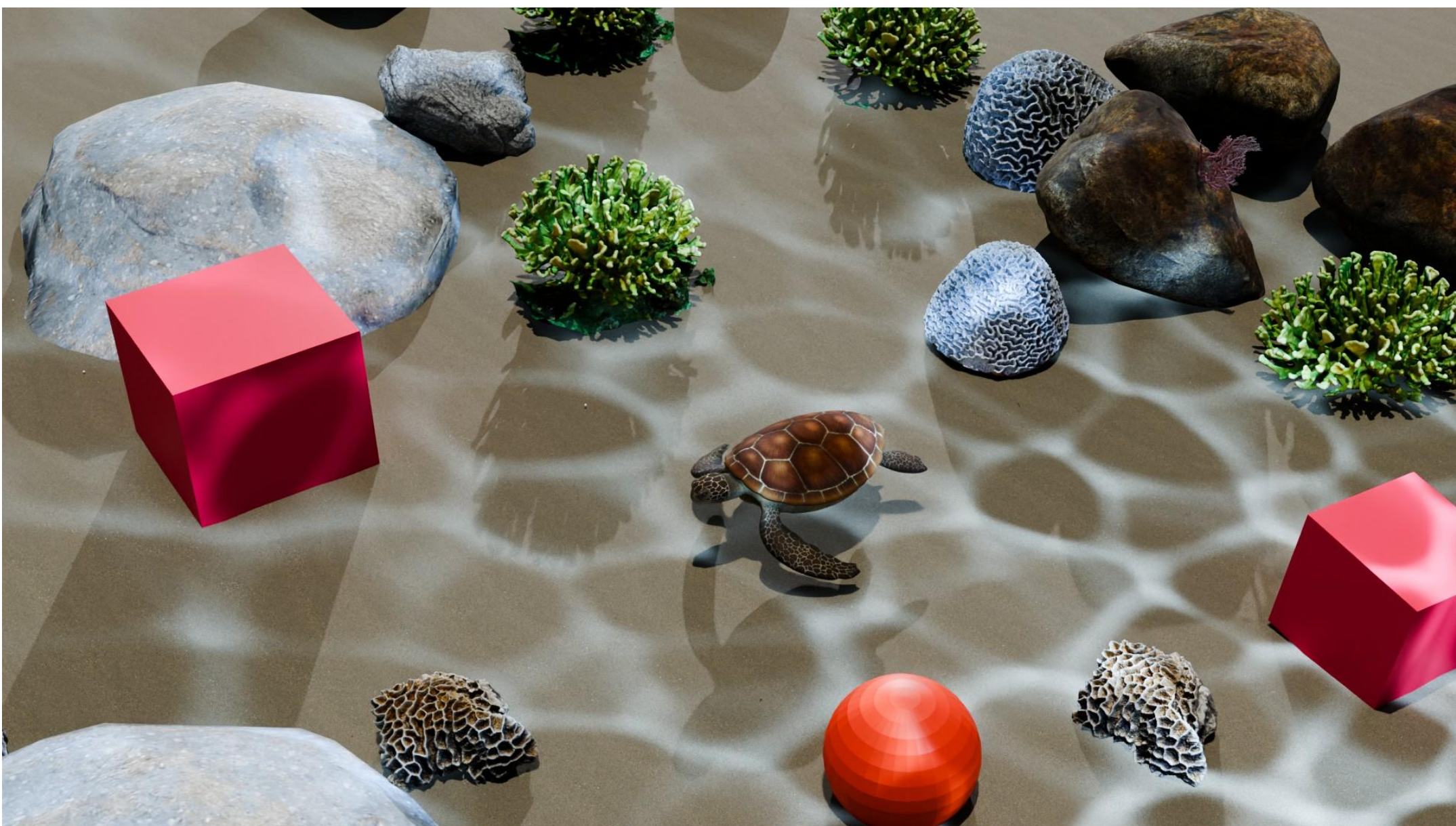


Figure 1. Example scene in Blender

Impact

Metric	Manual Diver Inspection	Project LiAIson (USV + ML)
Daily Cost	\$2,500 – \$5,000+	\$600 – \$1,200
Safety Risk	High (Life-threatening hazards)	Zero (Human remains on shore)
Review Time	Days (Manual playback/reporting)	Seconds (Automated flagging)
Throughput	Limited (Diver fatigue/air supply)	Continuous (24/7 capability)

Methodology

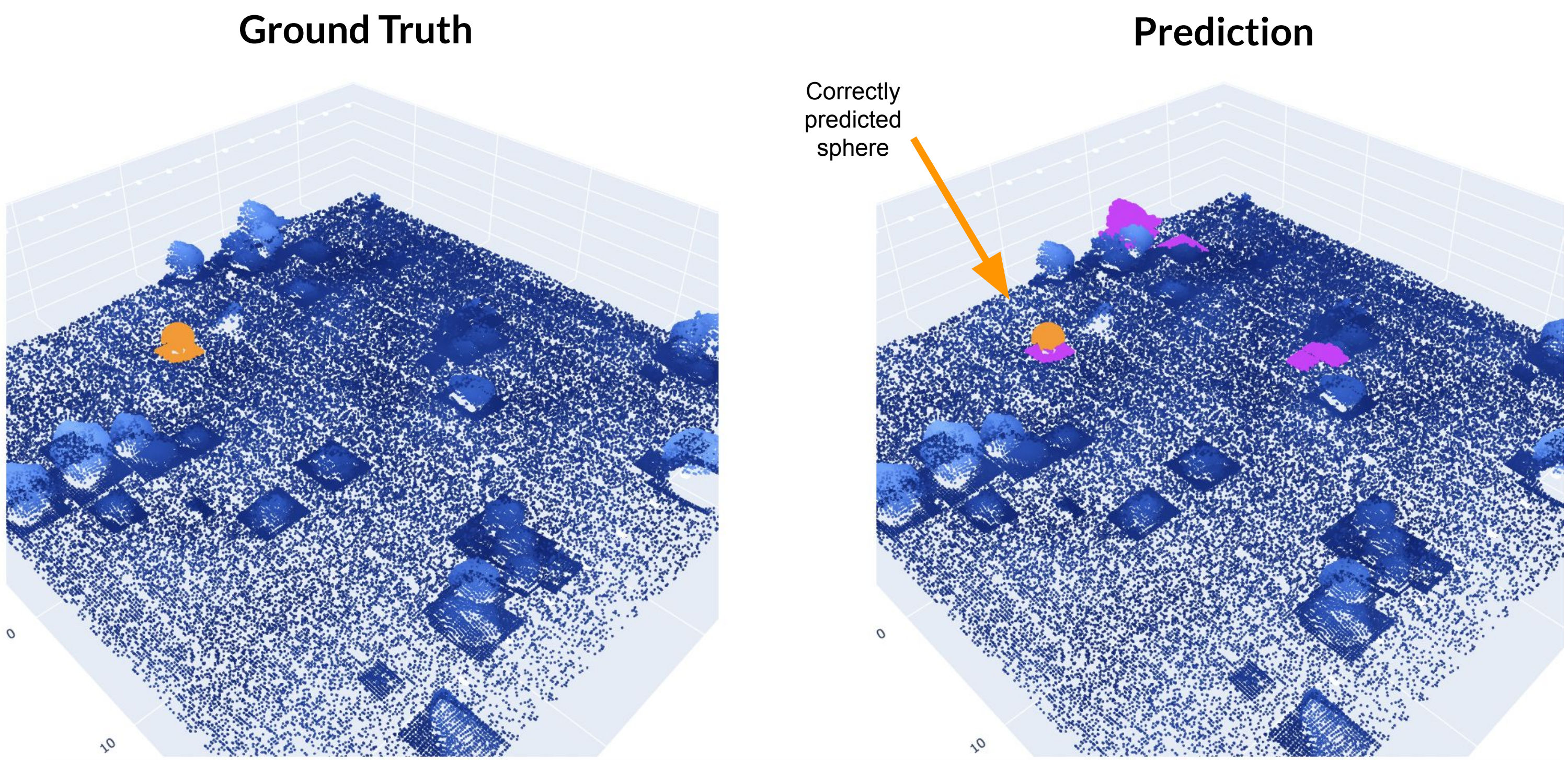
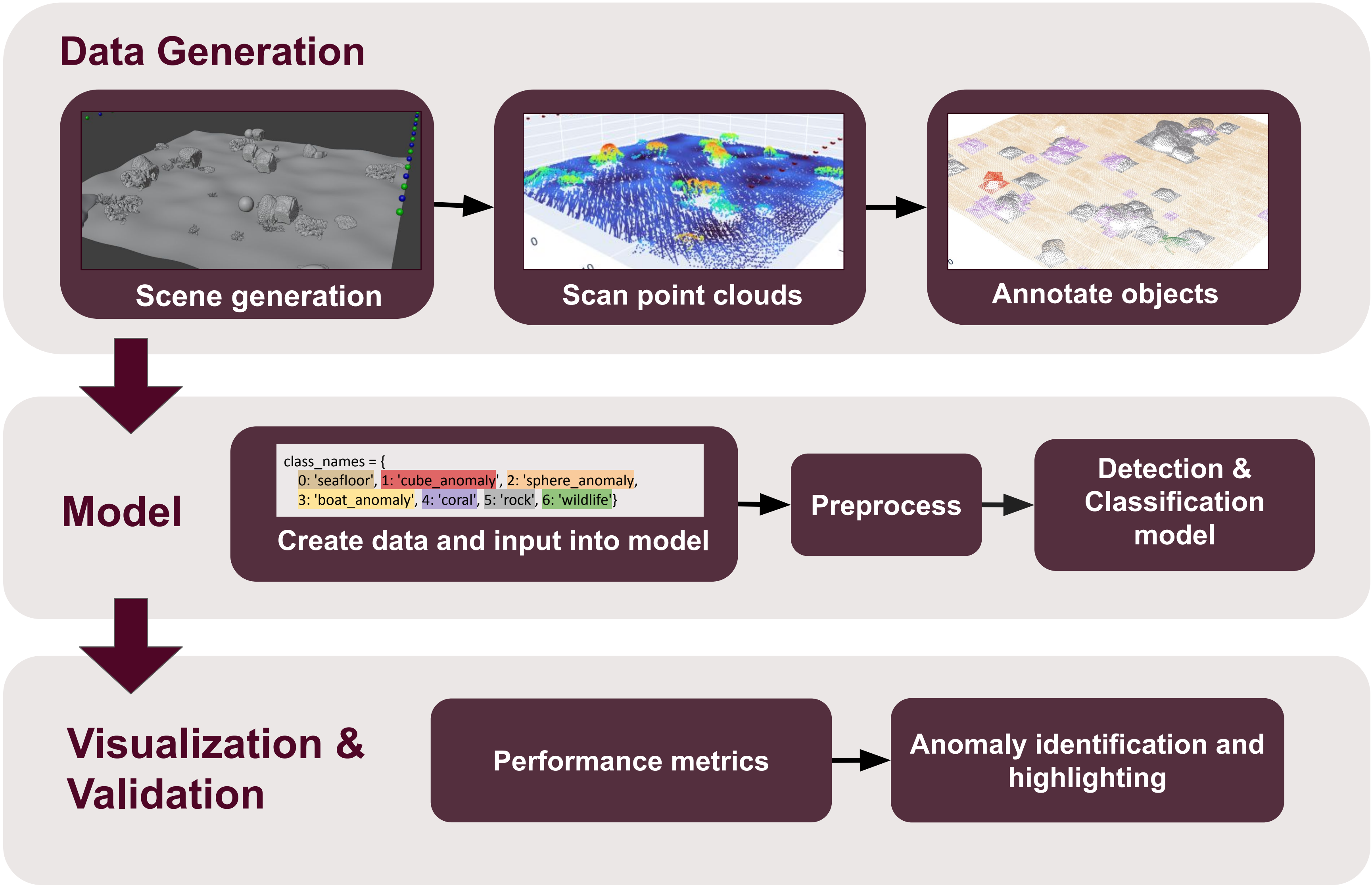


Figure 2. Ground truth versus predicted anomaly highlighting

~150 Synthetic Scenes

7 Classes

Results

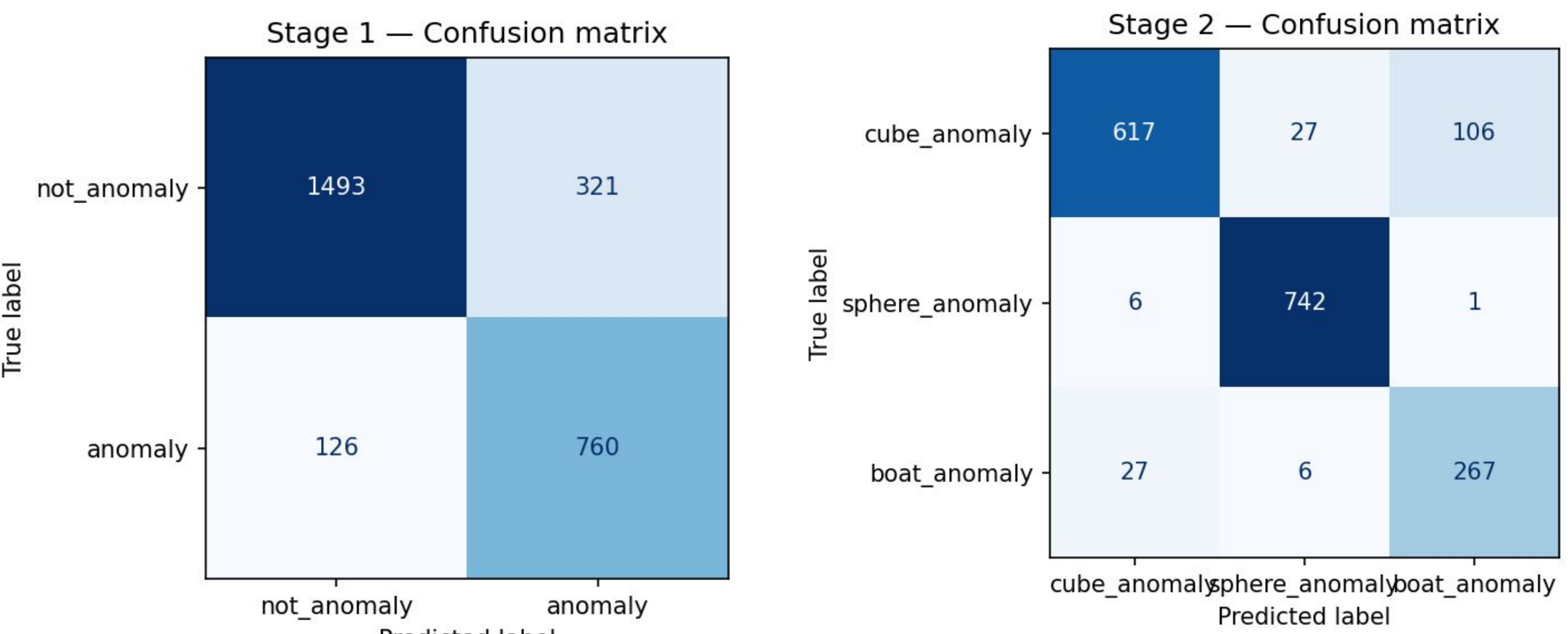
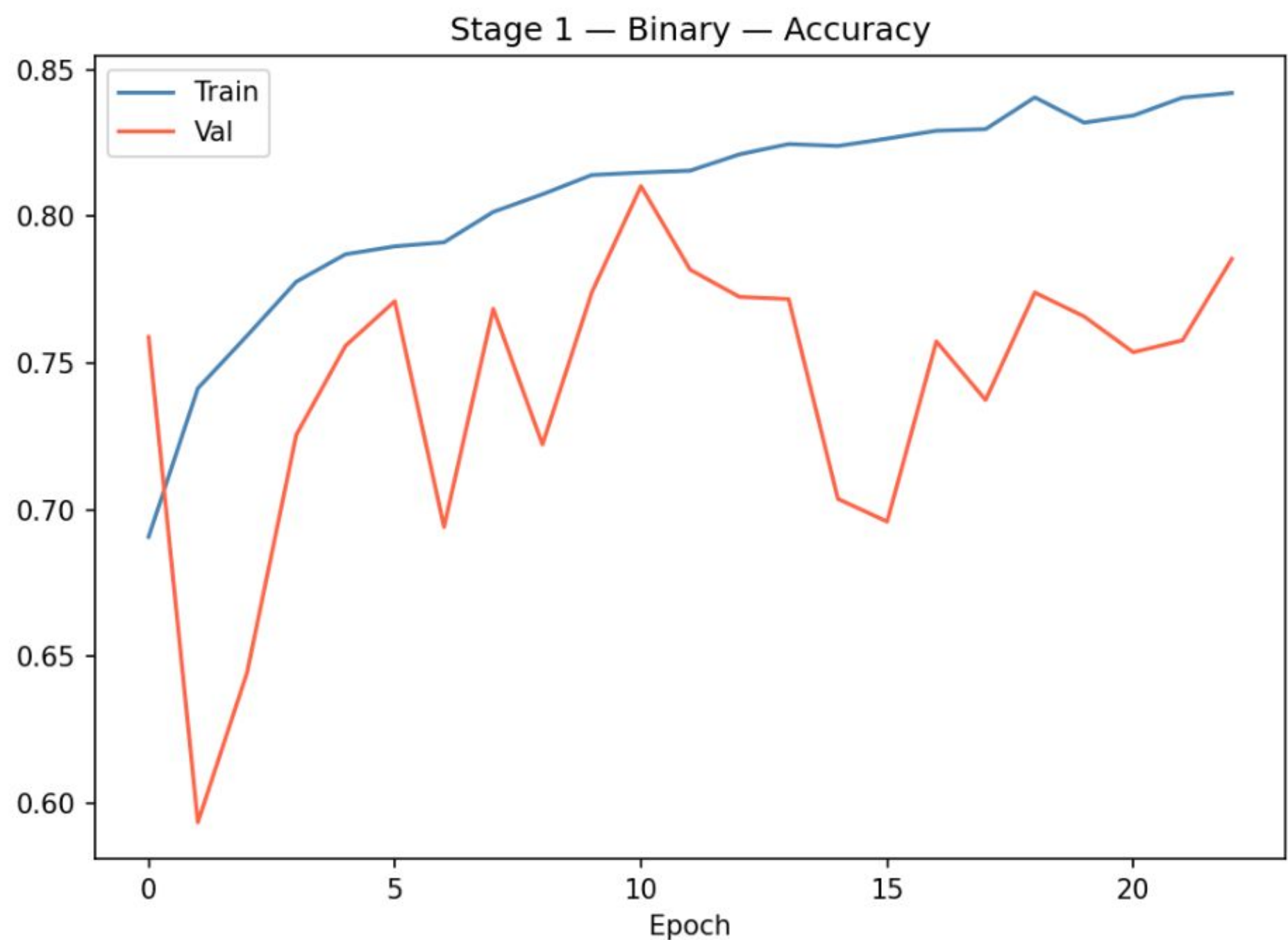


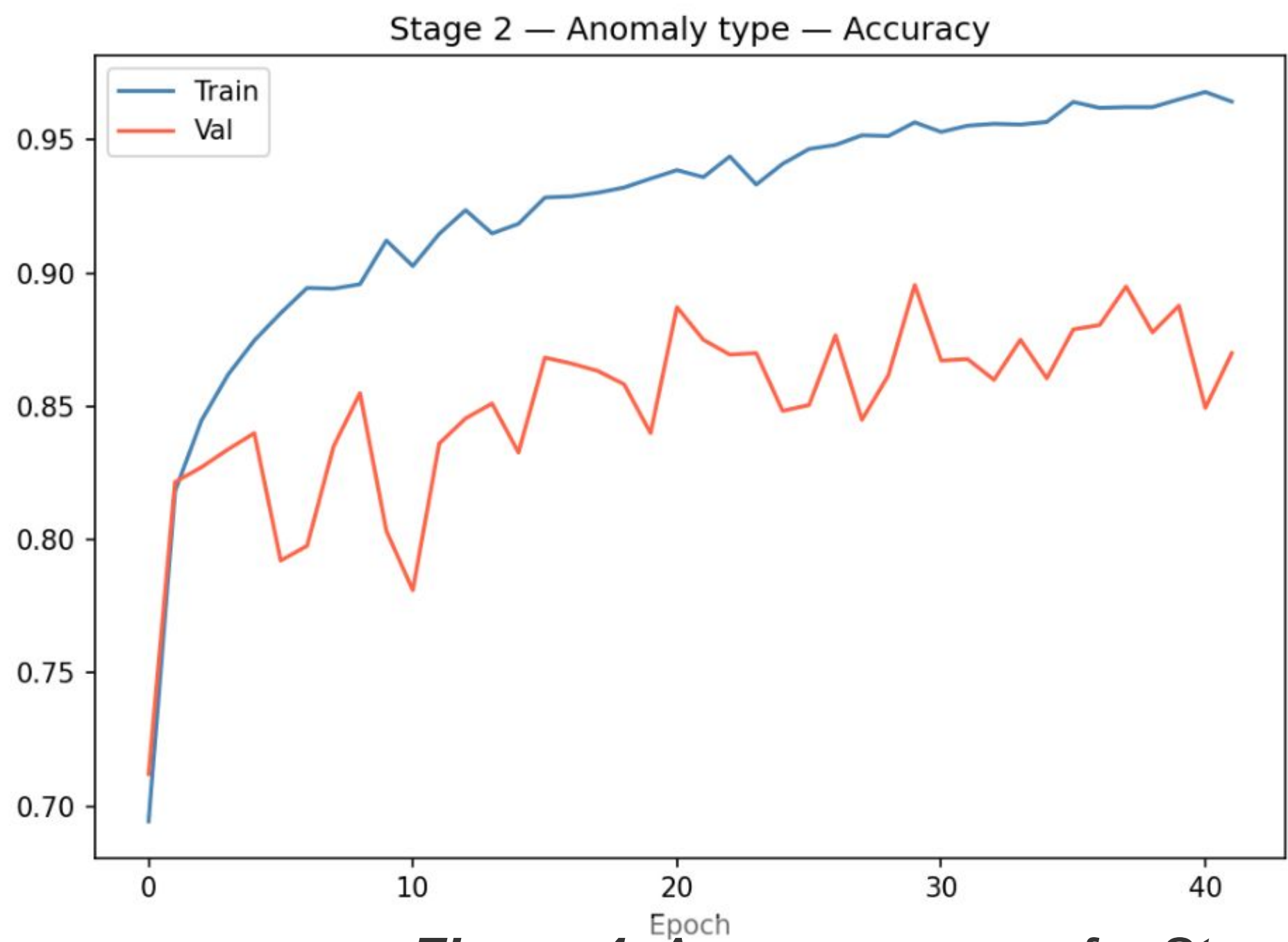
Figure 3. Confusion matrices for Stages 1 & 2



Accuracy
0.80

F1 Score
0.70

Precision
0.79



Accuracy
0.91

F1 Score
0.87

Precision
0.88

Figure 4. Accuracy curves for Stages 1 & 2

Conclusions

- Automated detection reduces inspection costs by 75% and increases data review throughput by 80%
- Synthetic pipeline enables safe, low-cost model training for new environments without expensive field trials.

References and Acknowledgements

1. Reitmunn S, Neumann L, Jung B. BLAInder-A Blender AI Add-On for Generation of Semantically Labeled Depth-Sensing Data. Sensors (Basel). 2021 Mar 18;21(6):2144. doi: 10.3390/s21062144. PMID: 33803908; PMCID: PMC8003152.

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